

Aerospace series

Plates

Aluminium alloy 7010/7050

Close-tolerance flatness

$6,0 \text{ mm} \leq a \leq 200,0 \text{ mm}$

Dimensions

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1 Scope

This standard specifies the dimensions, tolerances and mass of 7010/7050 plates with close tolerance flatness, thickness range $6,0 \leq a \leq 200,0$ mm for aerospace applications.

2 Normative references

This Airbus standard incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any these publications apply to this Airbus standard only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

AIMS 03-02-000	Technical specification – Aluminium and aluminium alloy wrought products – Plate
AIMS 03-02-022	Material specification – Aluminium alloy (7010/7050) solution treated, controlled stretched and artificially aged (T7451), Plate, $50,0 \text{ mm} \leq a \leq 200,0 \text{ mm}$, close tolerance flatness
AIMS 03-02-029	Material specification – Aluminium alloy (7010/7050) solution treated, controlled stretched and artificially aged (T7651), Plate, $6,0 \text{ mm} \leq a \leq 200,0 \text{ mm}$, close tolerance flatness

3 Form

See Figure 1.

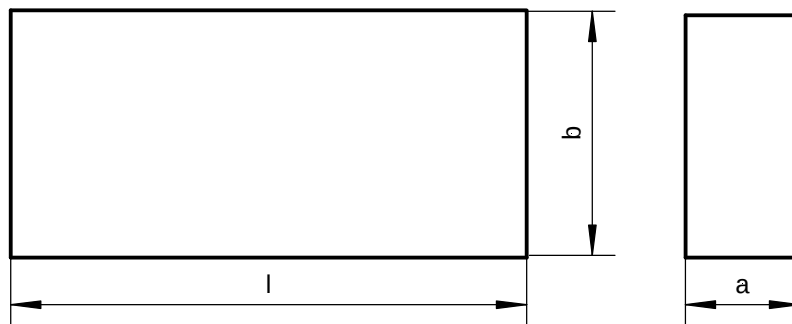


Figure 1:

4 Recommended dimensions

4.1 Dimensions and mass

See Table 1.

Table 1:

Size code	Nominal thickness a ¹⁾ mm	width x length b x l ¹⁾ mm x mm	Mass per unit area ²⁾ kg/m ²
006	6	1250 x 2500	16,8
008	8		22,4
010	10		28,0
012	12		33,6
014	14		39,2
016	16		44,8
020	20		56,0
025	25		70,0
030	30		84,0
035	35		98,0
040	40		112
045	45		126
050	50		140
055	55		154
060	60		168
065	65		182
070	70		196
080	80		224
090	90		252
100	100		280
110	110		308
120	120		336
130	130		364
140	140		392
150	150		420
160	160		448
170	170		476
180	180		504
190	190		532
200	200		560

1) Other thickness, width and length on request
2) For information, calculated with a density of 2,8 kg/dm³

5 Tolerances

5.1 Dimensional tolerances

5.1.1 Thickness

See Table 2.

Table 2:

Dimensions in mm

Thickness a	Thickness tolerances depending on plate thickness a and plate width b				
	b ≤ 1250	1250 < b ≤ 1600	1600 < b ≤ 2000	2000 < b ≤ 2500	2500 < b ≤ 3000
6 ≤ a ≤ 8	± 0,30	± 0,35	± 0,40	± 0,40	± 0,50
8 < a ≤ 10	± 0,35	± 0,40	± 0,40	± 0,45	± 0,55
10 < a ≤ 12	± 0,40	± 0,45	± 0,50	± 0,55	± 0,60
12 < a ≤ 16	± 0,50	± 0,55	± 0,60	± 0,65	± 0,70
16 < a ≤ 20	± 0,60	± 0,65	± 0,70	± 0,75	± 0,80
20 < a ≤ 25	± 0,70	± 0,75	± 0,75	± 0,85	± 0,85
25 < a ≤ 30	± 0,75	± 1,0	± 1,2	± 1,2	± 1,2
30 < a ≤ 35	± 0,85	± 1,1	± 1,3	± 1,3	± 1,3
35 < a ≤ 40	± 1,0	± 1,1	± 1,3	± 1,4	± 1,4
40 < a < 50	± 1,2	± 1,3	± 1,5	± 1,7	± 1,7
50 ≤ a ≤ 60	± 1,5	± 1,5	± 1,7	± 1,9	± 1,9
60 < a ≤ 70	± 1,7	± 1,8	± 2,0	± 2,2	± 2,2
70 < a ≤ 80	± 2,0	± 2,1	± 2,2	± 2,5	± 2,5
80 < a ≤ 100	± 2,2	± 2,2	± 2,4	± 2,7	± 2,7
100 < a ≤ 120	± 2,3	± 2,3	± 2,5	± 2,8	± 2,8
120 < a ≤ 140	± 2,5	± 2,5	± 2,7	± 3,0	± 3,0
140 < a ≤ 160	± 2,6	± 2,6	± 2,8	± 3,1	± 3,1
160 < a ≤ 180	± 2,7	± 2,7	± 2,9	± 3,2	± 3,2
180 < a ≤ 200	± 2,8	± 2,8	± 3,0	± 3,3	± 3,4

5.1.2 Width

See Table 3.

Table 3:

Dimensions in mm

Width b	Width tolerance
All	+10 0

5.1.3 Length

See Table 4.

Table 4: Dimensions in mm

Length l	Length tolerance
$l \leq 5000$	+10 0
$l > 5000$	+0,002 x l 0

5.2 Geometric tolerances

5.2.1 Squareness

5.2.1.1 Method of measurement and symbols

See AIMS 03-02-000.

5.2.1.2 Tolerances

See Table 5.

Table 5: Dimensions in mm

Length l	Maximum difference in lengths of diagonals for all plate widths and thickness
$l \leq 2000$	8
$2000 < l \leq 5000$	10
$l > 5000$	0,002 x l

5.2.2 Lateral curvature

5.2.2.1 Method of measurement and symbols

See AIMS 03-02-000.

5.2.2.2 Tolerances

See Table 6. The lateral curvature may be concave or convex.

Table 6: Dimensions in mm

Thickness a	Lateral curvature F on	
	plate width b	plate length l
$6 \leq a \leq 200$	$\leq 0,002 \times b$	$\leq 0,002 \times l$

5.2.3 Flatness

5.2.3.1 Method of measurement and symbols

See AIMS 03-02-000.

5.2.3.2 Tolerances

See Table 7.

Table 7:

Dimensions in mm

Plate thickness a	Flatness deviation $f_{\max}$. referred to					
	plate width b (T-bow)		plate length l (L-bow)		chord	
	width	$f_{\max}$	length	$f_{\max}$	chord	$f_{\max}$
$6 \leq a \leq 30$	$b \leq 500$ $b > 500$	1,5 $0,003 \times \min_1^{1)}$	$l \leq 750$	1,5	$w \geq 300$	$0,003 \times w^{3)}$
			$750 < l \leq 1000$	$0,002 \times l$		
			$1000 < l \leq 1300$	2,0		
			$l > 1300$	$0,0015 \times \min_2^{2)}$		
$30 < a \leq 200$	$b \leq 500$ $b > 500$	1,0 $0,002 \times \min_1^{1)}$	$l \leq 500$	1,0		
			$500 < l \leq 1000$	$0,002 \times l$		
			$1000 < l \leq 2000$	2,0		
			$l > 2000$	$0,001 \times \min_2^{2)}$		
1) $\min_1$ is either plate width (b) or straight-edge length, whatever is the smaller value 2) $\min_2$ is either plate length (l) or straight-edge length, whatever is the smaller value 3) The maximum acceptable deviation for total length and width shall not be exceeded.						

6 Material

See Table 8.

Table 8:

Plate thickness range mm	Material					
	AIMS spec. no.	delivered condition of treatment	code	AIMS spec. no.	use condition of treatment	code
$50 \leq a \leq 200$	03-02-022	T7451	A	03-02-022	T7451	A
$6 \leq a \leq 200$	03-02-029	T7651	B	03-02-029	T7651	B

7 Designation

Material according to this standard shall be designated following the philosophy of the example below:

	Description block	Identity block			
	Plate	<u>ABS5323</u>	<u>A</u>	<u>080</u>	<u>-01</u>
Number of ABS standard					
Material code (see table 8)					
Size code (see table 1)					
Blank if differentiation between 7010 / 7050 is not required					
Designation with alloy code only when required by design					
Code 01 for 7010					
Code 02 for 7050					

8 Technical specification

See AIMS03-02-000

RECORD OF REVISIONS

Issue	Clause modified	Description of modification
1 06/00	All	New standard
2 04/02	5.11 5.2.3.2	Table 2: thickness tolerances for width higher than 2000 mm have been added, Table 7: width range has been modified;
3 07/04	All	Inclusion of thickness range from 6 to 50 mm and AIMS 03-02-029 (T7651) on request of Airbus UK;
4 01/06	5.1.1	Table 2: width range $500 < b \leq 1250$ has been changed to $b \leq 1250$;
5 09/09	General	Use of latest template, AIMS 03-02-029 thickness range extended to $6,0\text{mm} \leq a \leq 200,0\text{mm}$;
6 12/09	2 7	Correction of AIMS 03-02-022 and AIMS 03-02-029 titles, Correction of size code and material code;
7 05/11	General 2 5.2.1.1, 5.2.2.1 5.2.3.1 5.2.1.2 Table 7 7	Editorial changes according to new template, table column headings reworded, EN 3848 removed, Methods of squareness, lateral curvature and flatness measurement refer now to AIMS 03-02-000 (before EN 3848), Squareness tolerance: maximum difference of diagonal lengths has only one limit, Limits revised based on flatness measurement method defined in AIMS 03-02-000 and lower flatness limits introduced, footnotes added, Wording revised;